BANGALORE UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING, UVCE, BENGALURU
B.Tech. PROGRAMME IN COMPUTER SCIENCE AND ENGINEERING

Course Code	**18CIPC801**					
Category	Engineering Science Courses : Professional Core					
Course title	**DATA SCIENCE - THEORY**					
Scheme and Credits	No. of Hours/Week					Semester - VIII CSE/ISE
	L	T	P	SS	Credits	
	2	2	0	0	3	
CIE Marks: 50	SEE Marks: 50		Total Max. Marks: 100			Duration of SEE: 03 Hours
Prerequisites (if any): NIL						

COURSE OBJECTIVES:

This course will enable students to

1. Understand the principles of Data warehousing and data mining.
2. Build and Develop a Data Warehouse and mapping the data warehouse to a multiprocessor architecture.
3. Perform classification and prediction of data.
4. Examine the types of data in cluster analysis with various Clustering techniques.
5. Know the various Data analytics and Data visualization techniques.

UNIT I: INTRODUCTION TO DATA MINING **10 Hours**

Importance of Data Mining, Data Mining Functionalities, Different kinds of Data, Classification of Data Mining Systems, Primitives, Major Issues and challenges in Data Mining, Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization and Concept Hierarchy Generation.

UNIT II: DATA WAREHOUSING **09 Hours**

Data warehousing Components, Building a Data warehouse, Mapping the Data Warehouse to a Multiprocessor Architecture, DBMS Schemas for Decision Support, Data Extraction, Cleanup and Transformation Tools, Metadata, reporting, Query tools and Applications, Online Analytical Processing (OLAP), OLAP and Multidimensional Data Analysis.

UNIT III: CLASSIFICATION AND PREDICTION **09 Hours**

Supervised and unsupervised learning, Definition of classification, Decision Tree Induction, Bayesian Belief networks, Bayesian classification, Rule Based Classification, K-nearest neighbor method, Prediction using Linear and Non-linear Regression, Classification Accuracy and Error Measures.

UNIT IV: CLUSTERING AND ASSOCIATION RULE MINING **10 Hours**

Types of Data in Cluster Analysis, Classification of clustering methods, K-means, BIRCH, DBSCAN, STING Algorithms, Outlier Analysis. Basic concepts of Association Rules, Classification of Association Rules, Apriori Algorithms, FP Tree, Multilevel Association rules, Categorical Association Rules, Multidimensional Association Rules. Mining Frequent closed Itemsets, Metarule-guided Association Rules, Constraint Based Association Rules.

UNIT V: DATA ANALYTICS AND VISUALIZATION **10 Hours**

Introduction to Data Analysis, Applied Statistical techniques, Types of Statistical Data, Types of Big Data Analytics, Collecting data for Sampling and Distribution, Probability, Frequency Distribution, Population and Parameters, Central Tendency, Measures of Central Tendency, Different Types of Statistical Means, Problems of Estimation and Normal Distribution curve. Data Visualization: Basic Principles, Importance, Conventional Data Visualization Methods, Ideas and tools for Data Visualization.

TEXT BOOKS:

1. Jiawei Han and Micheline Kamber, Data Mining, Concepts and Techniques, Elsevier, Third Edition, 2011.
2. Pang-Ning Tan, Michael Steinbach and Vipin Kumar, Data Mining Algorithms, 2005.

REFERENCE BOOKS:

1. V K Jain, Data Science and Analytics, Khanna Publishing, 2018.
2. David Hand, Heikki Mannila, Padhraic Smyth, Principles of Data Mining, PHI, 2009.
3. Margaret H Dunham, Data Mining Introductory and Advanced Topics, Pearson Education, 2008.

e-BOOKS/ONLINE RESOURCES:

1. http://en.wikipedia.org/wiki/Data_warehouse
2. http://www.inf.unibz.it/dis/teaching/DWDM/slides/dw1.pdf
3. https://repo.palkeo.com/algo/informationretrieval/Data%20mining%20and%20analysis.pdf

MOOCs:

1. https://www.coursera.org/specializations/jhu-data-science?siteID=OyHlmBp2G0c-0328ZKV34mF3.yMgOBpdWA&utm_content=2&utm_medium=partners&utm_source=linkshare&utm_campaign=OyHlmBp2G0c.
2. https://www.thisismetis.com/courses/introduction-to-data-science?utm_source=LDS&utm_medium=affiliate&utm_campaign=LDS2019affiliate
3. https://www.udemy.com/python-for-data-science-and-machine-learning-bootcamp/?ranMID=39197&ranEAID=OyHlmBp2G0c&ranSiteID=OyHlmBp2G0c-wgJMi8qQiA2u1hpioHWhbQ&LSNPUBID=OyHlmBp2G0c
4. https://www.datacamp.com/?tap_a=5644-dce66f&tap_s=97692-82206a

COURSE OUTCOMES:

The students at the end of the course, will be able to

CO1: Demonstrate the concept of data mining principles.

CO2: Discuss the Data Warehousing Architectures and its Implementation.

CO3: Apply the association rules, design and deploy appropriate classification techniques for mining the data.

CO4: Cluster the high dimensional data for better organization of the data.

CO5: Explore the knowledge of various techniques of data analytics and data visualization Tools.

SCHEME OF EXAMINATION:

<table>
<tr><td rowspan="2">CIE – 50 Marks</td><td>Test I (Any Three Units) - 20 Marks</td><td>Quiz I – 5 Marks</td><td>25 Marks</td><td rowspan="2">Total: 50 Marks</td></tr>
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BANGALORE UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING, UVCE, BENGALURU

B.Tech. PROGRAMME IN COMPUTER SCIENCE AND ENGINEERING

<table>
<tr><td>Course Code</td><td colspan="6">18MCIP803</td></tr>
<tr><td>Category</td><td colspan="6">Mandatory Course</td></tr>
<tr><td>Course title</td><td colspan="6">INTELLECTUAL PROPERTY RIGHTS</td></tr>
<tr><td rowspan="3">Scheme and Credits</td><td colspan="5">No. of Hours/Week</td><td rowspan="3">Semester - VIII CSE/ISE</td></tr>
<tr><td>L</td><td>T</td><td>P</td><td>SS</td><td>Credits</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>CIE Marks: 50</td><td colspan="2">----</td><td colspan="2">Total Max. Marks: 50</td><td colspan="2"></td></tr>
<tr><td colspan="7">Prerequisites (if any): NIL</td></tr>
</table>

COURSE OBJECTIVES:

The course will enable the students to

1. Acquire the main objective of the course paper IPR is to make the students aware of their rights for the protection of their invention done in their project work.
2. Get registration in our country and foreign countries of their invention, designs and thesis or theory written by the students during their project work and for this they must have knowledge of patents, copy right, tradeMarks, designs.
3. Know about the registration of IPR for adding credit to their work done.
4. Generate students to build their career as entrepreneurs for eradicating unemployment with the concerned field.
5. Bring in interest amongst the students for generating thinking and for igniting the young minds for bringing new ideas and research in the field of technology.

UNIT I: INTRODUCTION **3 Hours**

Meaning of Property, Origin, Nature, Meaning of Intellectual Property Rights, Provision of IPR under TRIPS and WTO. Kinds of Intellectual property rights—Copyright, Patent, Trade Mark, Trade Secret and trade dress, Design, Layout Design, Geographical Indication, Plant Varieties and Traditional Knowledge.

UNIT II: PATENT RIGHTS AND COPY RIGHTS **10 Hours**

Origin, Meaning of Patent, Types, Inventions which are not patentable, Registration Procedure, Rights and Duties of Patentee, Assignment and licence, Restoration of lapsed Patents, Surrender and Revocation of Patents, Infringement, Remedies & Penalties. **COPY RIGHT:** Origin, Definition &Types of Copy Right, Registration procedure, Assignment & licence, Terms of Copy Right, Infringement, Remedies, Copyrights with special reference to software.

UNIT III: TRADE MARKS AND DESIGNS **7 Hours**

Origin, Meaning & Nature of Trade Marks, Types, Registration of Trade Marks, Infringement & Remedies, Offences relating to Trade Marks, Passing Off, Penalties. **DESIGN:** Meaning, Definition, Object, Registration of Design, Cancellation of Registration, International convention of design- types and functions. Semiconductor Integrated circuits and layout design Act-2000.

UNIT IV: INNOVATION **3 Hours**

Indicators for Innovation – Innovation for improvement of business keeping in view stakeholders expectation; Leadership Indication – Business innovation in products, process and methods of stakeholders engagement for the purpose of overall improvement of the business; Pardoning mistakes by employees in business and encouraging for experimentation for innovative ideas; Rewarding for good experimentation and innovation; exploring and absorbing cutting edge technology; Pay for innovative results; Budget for Research and Development;

UNIT V: ENTREPRENEURSHIP **3 Hours**

Community Based Micro Finance Institute (CBMFI) by choosing self-group having overall good track record for creation of entrepreneurship; Entrepreneurship through Community Micro Vendor Development Programme (CMVDP) by providing proper training and skills to generate business in village areas and act as supply chain for companies around the village by way supplying products; Entrepreneurship skills, Market Place, Initiating a venture, planning for a venture and evaluating a venture; Entrepreneurship for women.

REFERENCE BOOKS:

1. Cornsih W.R. Intellectual Property, Patents, TradeMarks, Copyrights and Allied Rights (1999), Asia Law House, Hyderabad.
2. Intellectual Property Rights and the Law, Dr. G.B. Reddy, Gogia Law Agency.
3. Law relating to Intellectual Property, Dr. B. L. Wadehra, Universal Law Publishing Co.
4. P Narayan, Intellectual Property Law (1999), (ed.) Eastern Law House, Calcutta.
5. Bibeck Debroy (ed) Intellectual Property Rights (1998), Rajiv Gandhi Foundation, Delhi.
6. Entrepreneurship and Innovation: Readings and Cases Paperback – April 15, 2011 by Tim Mazzarol (Author)
7. Corporate Social Responsibility, C A Kamal Garg, 2014, BHARAT LAW HOUSE Pvt. Ltd., New Delhi.
8. Peter F. Drucker, Harper Collins, 17-Mar-2009 - Business & Economics - Classic book on Innovation and Entrepreneurship
9. Innovation and Entrepreneurship: A Competency Framework by Charles H. Matthews, Ralph Brueggemann, 2015, Routledge.

COURSE OUTCOMES:

The students at the end of the course, will be able to

CO1: Get awareness of acquiring the patent and copyright for their innovative works.

CO2: They also get the knowledge of plagiarism in their innovations which can be questioned legally.

CO3: Apply the acquired concepts of Innovation and Entrepreneurship in their respective areas.

CO4: The yet another important outcome of the course is to make the students as leaders in the area of IPR so that their rights are not infringed in society.

CO5: In recent times Copyright is undergoing changes in the emerging trend of copying through internet and other sources, therefore the outcome of the course is to end this type of unwarranted copying from different medias.

SCHEME OF EXAMINATION:

CIE – 50 Marks	Test I (UNIT I, II & III) - 20 Marks	Quiz I – 5 Marks	25 Marks	Total: 50 Marks
	Test II (UNIT IV & V) - 20 Marks	Quiz II – 5 Marks	25 Marks	

Note: SEE shall be conducted for 100 Marks and the Marks obtained is scaled down to 50 Marks.

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COURSE OBJECTIVES:

Course Code	**18CIPC804**					
Category	Engineering Science Courses : Professional Core					
Course title	**PROJECT WORK**					
Scheme and Credits	No. of Hours/Week					Semester – VIII CSE/ISE
	L	T	P	SS	Credits	
	0	0	18	0	09	
CIE Marks: 50	SEE Marks: 50		Total Max. Marks: 100			Duration of SEE: 03 Hours
Prerequisites (if any): NIL						

The course will enable the students to

1. Apply / use different experimental techniques, experiments, computational or analytical modelling and simulation tools to carry out tests and generate the results.
2. Analyze the results of experiments conducted / models developed.
3. Create a detailed technical document in prescribed format on the outcome of preliminary project and project work.
4. Prepare a technical presentation to the Project Evaluation Committee of the Department.

GUIDELINES:

1. Project work is a continuation of preliminary project work started in 7th semester.
2. Project team has to execute the work and test the prototype / algorithm within the timeline.
3. Project team has to demonstrate the successful working of prototype / algorithm developed.
4. Project team has to document the work carried out in the form of a report in prescribed format and submit to the department.
5. Project team has to make a technical presentation of the work carried out to the Project Evaluation Committee of the Department.

COURSE OUTCOMES:

The students at the end of the course, will be able to

CO1: Apply the engineering principles in planning, formulating an innovative design/approach to problem solving.

CO2: Develop/implement the design with appropriate techniques, resources and contemporary tools.

CO3: Plan, monitor and manage project schedule, resources and work assignments to ensure timely completion.

CO4: Test and evaluate the performance of the implemented project and understand the significance of the solution.

CO5: Perform professionally as a team member, accept responsibility, take initiative and provide leadership necessary to ensure project success.

CO6: Use formal and informal communications with team members and guide, make presentations and prepare technical documents.

CO7: Provide solution within the legal framework addressing societal and environmental concerns and exhibit integrity and ethical behaviour in engineering practice.

SCHEME OF EXAMINATION

Rubrics for CIE:

1. Introduction and justification of topic : 10%
2. Literature survey and conclusion : 20%
3. Objectives and scope of project work : 10%
4. Methodology to be adopted : 30%
5. Presentation of contents of project work : 30%

Rubrics for SEE:

1. Introduction and justification of topic : 10%
2. Literature survey and conclusion : 20%
3. Objectives and scope of dissertation work : 10%
4. Methodology, Experimental / software : 30%
5. Presentation of project work : 30%

NOTE: The percentage mentioned above indicates marks allocation.

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BANGALORE UNIVERSITY

Course Code	**18CIIN805**					
Category	Engineering Science Courses : Mandatory Course					
Course title	**INTERNSHIP**					
Scheme and Credits	No. of Hours/Week					Semester – VIII CSE/ISE
	L	T	P	SS	Credits	
	0	0	6	0	3	
CIE Marks: 100			Total Max. Marks: 100			
Prerequisites (if any): NIL						

Importance:

- Internships are educational and career development opportunities, providing practical experience in a field or discipline.
- The main aim is enhancement of the employability skills of the students passing out from technical institutions.
- Internships can be full time during vacations and part time during academic session.

1. Benefits to Students:

a) Opportunity to see how the theoretical aspects learnt in classrooms are integrated into the practical world.
b) Opportunity to learn –
   i. New skills and supplement acquired knowledge.
   ii. Teamwork skills.
   iii. Time management, multi-tasking in an industrial environment.
c) Helps them decide to choose the industry and the career profile.
d) Provides opportunity to evaluate the organization before committing to a full time position.

2. Benefits to Institute:

a) Build meaningful Industry-Institutional interaction and makes the placement process easier.
b) Curriculum revision can be made based on feedback from Industry/ students.
c) Improve institutional credibility & branding.

3. Benefits to Industry:

(a) Availability of ready to contribute candidates for employment.
(b) New perspectives to problem solving by students undergoing internship.
(c) Availability of quality candidates for temporary or seasonal positions in ongoing projects.

4. Scheme of Evaluation:

Internship activities and allocation of marks for credit computation are listed in Table 1. Students shall carryout internship full time during vacation and/or part time during academic session, over a period of 3 years starting from II Sem vacation and the evaluation shall be carried out during VIII Sem B.Tech. examination.

Students may choose any number of activities as per their interest and based on their convenience and earn internship activity marks and credits.

A student shall earn a total of 100 marks and 3 credits from the activities mentioned in Table 1. Internship shall be evaluated by the Proctor, based on the report of activities performed.

Note: Marks allocated for performance in internship activity are as under: Excellent – 20, Good – 15, Satisfactory – 10

If the certificate does not indicate any performance level, it shall be treated as excellent.

Table 1: Internship activities and allocation of marks for credit computation.

Sl. No.	Activity Head	Document as evidence	Evaluated by	Performance appraisal	Max. Marks
1	Inter/Intra institutional /Workshop/ training	Certificate	Programme Coordinator	Satisfactory/Good/ Excellent	Participation: 10 Organisation: 10
2	Working for consultancy/research project	Certificate	Principal Investigator	Satisfactory/Good/ Excellent	20
3	Festival (Technical, Business, Cultural & others)	Certificate	Faculty In-charge / Principal	Satisfactory/Good/ Excellent	Participation: 10 Organisation: 10
4	Contribution in Incubation/ Innovation / Entrepreneurship cell/ Institutional Innovation Council	Certificate	Cell In-charge	Satisfactory/Good/ Excellent	20
5	Participation in innovation related competition – Hackathons etc	Certificate	Faculty Proctor	Satisfactory/Good/ Excellent	20
6	Development of new product/business plan/registration of start-up	Certificate	Faculty In-charge	Satisfactory/Good/ Excellent	40
7	Participation in all activities of Institute Ex. IPR workshop, leadership task, idea, design, innovation, business competition,	Certificate	Faculty Coordinator / Principal	Satisfactory/Good/ Excellent	Participation: 5 Organisation: 5 5 marks for each activity subject to max. of 20.
8	Work experience at family business	Declaration by parent	TPO	Satisfactory/Good/ Excellent	20
9	Internship with industry, Govt., NGO, PSU, MSME, Online internship	Detailed report	Faculty Proctor / TPO / Industry Supervisor	Satisfactory/Good/ Excellent	20-40 2 weeks: 20 4 weeks: 30 6 weeks: 40
10	Rural internship	Detailed report & Certificate	Faculty Proctor/ TPO/NCC/NSS Head	Satisfactory/Good/ Excellent	20 (any one activity)

The following are rural internship activities that may be carried out by students in teams:

1. Prepare and implement plan to create local job opportunities.
2. Prepare and implement plan to improve education quality in village.
3. Prepare an actionable Detailed Project Report for Doubling the village Income.
4. Developing Sustainable Water Management system.
5. Prepare and Improve a plan to improve health parameters of villagers.
6. Developing and implementing Low Cost Sanitation facilities.
7. Prepare and implement plan to promote Local Tourism through Innovative Approaches.
8. Implement/Develop technology solutions which will improve quality of life.
9. Prepare and implement solution for energy conservation.
10. Prepare and implement plan to skill village youth and provide employment.
11. Develop localized techniques for reduction in construction Cost.
12. Prepare and implement plan for sustainable growth of village.
13. Setting of Information imparting club for women leading to contribution in social and economic issues.
14. Developing and managing efficient garbage/ solid waste disposable system.
15. Contribution to any national level initiative of Government of India. For eg. Digital India/ Skill India/ Swachh Bharat Internship, Sansad Adarsh Gram Yojana, etc.

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B.Tech. PROGRAMME IN COMPUTER SCIENCE AND ENGINEERING

Course Code	**18CIPE86B**					
Category	Engineering Science Courses : Professional Elective					
Course title	**GREEN COMPUTING – THEORY**					
Scheme and Credits	No. of Hours/Week					Semester - VIII CSE/ISE
	L	T	P	SS	Credits	
	2	2	0	0	3	
CIE Marks: 50	SEE Marks: 50		Total Max. Marks: 100			Duration of SEE: 03 Hours
Prerequisites (if any): NIL						

COURSE OBJECTIVES:

The course will enable the students to

1. Learn what Green IT is and how it can help to improve environmental Sustainability.
2. Apply the concepts related to Green devices and hardware along with software techniques.
3. Analyse the Green enterprise activities, managing the green IT is adopted or deployed in enterprises.
4. Evaluate various laws, standards, and protocols.
5. Implement the strategies to develop Green IT.

UNIT I: GREEN IT **09 Hours**

Environmental Concerns and Sustainable Development, Environmental Impacts of IT, Green IT, Holistic Approach, Enhancing Environmental sustainability, IT Standards and Eco-Labelling of IT, Enterprise IT strategy, Merits and Demerits.

UNIT II: GREEN HARDWARE WITH GREEN SOFTWARE **10 Hours**

Green Hardware: Life Cycle of a device or hardware, Reuse, Recycle and Dispose. Green Software: Energy-saving software techniques, evaluating and Measuring Software Impact to platform power.

UNIT III: GREEN ENTERPRISES AND THE ROLE OF IT **10 Hours**

Organization and Enterprise, Greening Information Systems, Greening the Enterprise: IT Usage and Hardware, Inter-Organizational Enterprise activities and Issues, Enablers and making the case for IT and Green Enterprise.

UNIT IV: MANAGING & REGULATING THE GREEN IT **10 Hours**

Managing Green IT: Strategizing Green Initiatives, Implementation, Information Assurance, Communication and Social media. Regulating the Green IT: Laws, Standards and Protocols: The regulatory environment and IT manufacturers, Non regulatory government initiatives, Industry associations and standard bodies, Building standards, Data centres, Social movements and Greenpeace.

UNIT V: IMPLEMENTATION **09 Hours**

Awareness to implementations, Greening by IT, A megatrend, A creation of green IT strategy, Research and Development directions.

TEXT BOOKS:

1. San Murugesan, G. R. Gangadharan: Harnessing Green IT, WILEY -2013.
2. Bhuvan Unhelkar: Green IT Strategies and Applications: Using Environmental Intelligence (Advanced & Emerging Communications Technologies) – 2011.

REFERENCE BOOKS:

1. Alin Gales, Michael Schaefer, Mike Ebbers, "Green Data Center: Steps for the Journey", Shoff/IBM Rebook, 2011.
2. John Lamb, "The Greening of IT", Pearson Education, 2009.
3. Carl speshocky, "Empowering Green Initiatives with IT", John Wiley & Sons, 2010.
4. Wu Chun Feng (editor), "Green computing: Large Scale energy efficiency", CRC Press, 2012.

e-BOOKS /ONLINE RESOURCES:

1. http://shodh.inflibnet.ac.in:8080/jspui/bitstream/123456789/424/2/02_introduction.pdf
2. http://www.greencompliance.com
3. https://www-03.ibm.com/press/attachments/GreenIT-final-Mar.4.pdf
4. https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118305393.ch2

MOOCs:

1. https://www.apus.edu/schedule-classes/schedule/course/issc387
2. https://www.researchgate.net/publication/221939724_Need_of_Green_Computing_Measures_for_Indian_IT_Industry
3. https://www.greenit.net/greenit_training.html
4. https://www.slideshare.net/ranjanagore/enterprise-green-it-strategy

COURSE OUTCOMES:

The students at the end of the course, will be able to

CO1: Understand Green IT with its different dimensions to the Environmental Sustainability.
CO2: Compare Green devices and hardware along with its green software methodologies.
CO3: Analyse the various Green enterprise activities, functions and their role with IT.
CO4: Design Data centres with standard laws and protocols for regulating Green IT.
CO5: Validate the key strategies for Sustainability, Software Impact to platform power & Regulatory/Non-regulatory government initiatives.

SCHEME OF EXAMINATION:

<table>
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